

Buck Converter Problems

1.

The circuit shown in Figure 1 is a buck converter. It is stepping from an input voltage of 12 volts (from a battery some distance away) to an output voltage of 5 volts, with an average load current of 0.6 amp. It is switching at 100kHz.

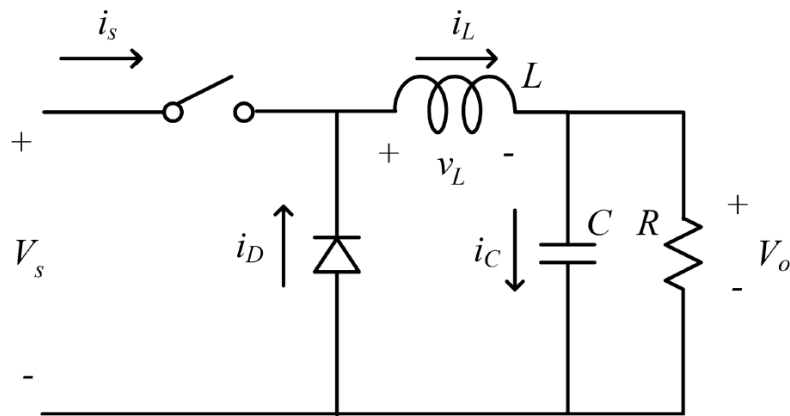


Figure 1: Buck converter

- Assuming the inductor energy storage is continuous by a reasonable margin, sketch waveforms for the circuit, showing the inductor voltage v_L , the inductor current i_L , the supply current i_s , the capacitor current i_C , and the expected capacitor voltage ripple. Make sure the timing is marked clearly.
- Derive an expression for the step-down ratio of the converter as a function of the duty cycle D .
- Calculate the minimum inductance needed to maintain continuous inductor current.
- This circuit needs an input capacitor to manage the rapidly changing input current i_s . Assuming that the inductor current ripple is 0.2A peak to peak, calculate the capacitance value to keep the input ripple voltage to less than 0.5 volts peak to peak.

2.

Draw the two equivalent circuits of a buck converter (switch on, and switch off) with the controllable switch in the low-side position. Also explain why buck converters are often designed to have the switch in the low-side position.

3.

Explain why a dedicated MOSFET gate-driver circuit is needed for high switching frequency buck converters, and identify the characteristics that are needed for a good gate-driver circuit.